MULTI-AGENT ELECTION LAB

Who wins depends on who you hear

A simulated town of 300 voters. We change one condition at a time and watch who wins.

Team Eclipse · 9 November 2026

THE QUESTION

Does the winner depend only on what people prefer, or also on how they reason and what they can see?

EXPERIMENT 1

Everyone sees everything

Full information · 30 towns · 40 rounds

EXPERIMENT 2

How voters reason

84 voter mixes × 30 towns

EXPERIMENT 3

Who they can hear

5 hearing radii, walking or still, × 30 towns

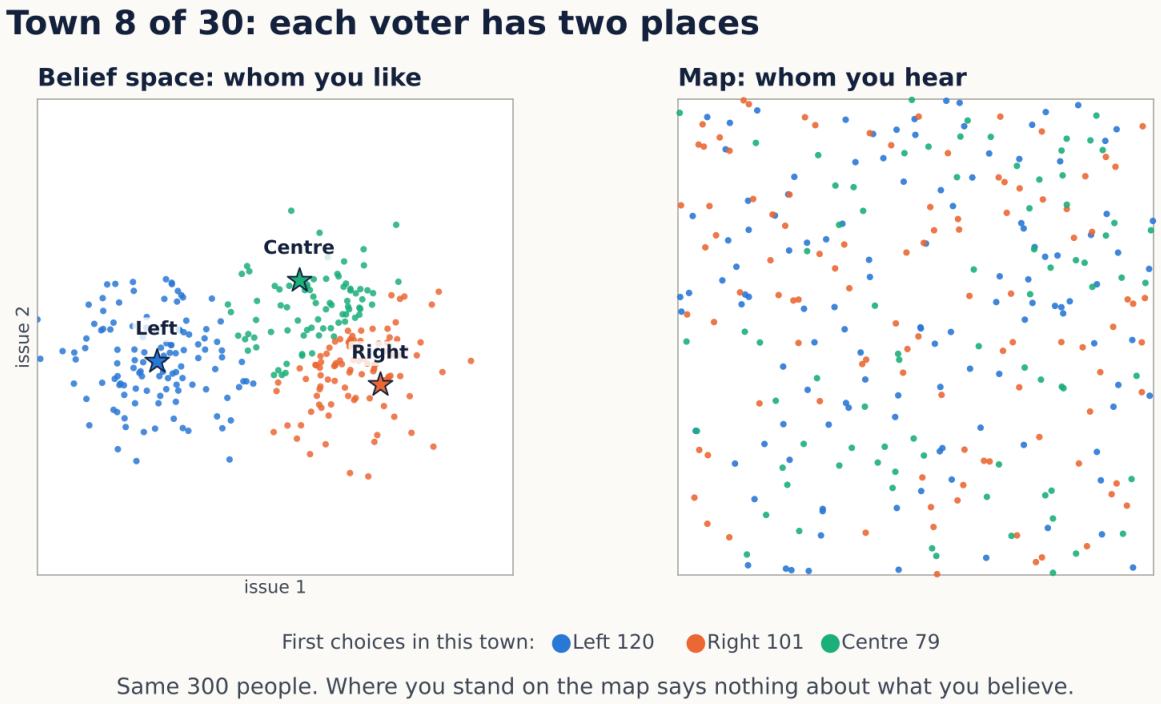
EXPERIMENT 4

Who tries to sway them

A loyal 10% with 0 to 80 followers × 30 towns

THE TOWN

300 voters, each in two places



Town 8 of 30. First choices: Left 120, Right 101, Centre 79. Each dot keeps its colour, its favourite, as it moves to a random spot on the map.

BELIEF SPACE: WHOM YOU LIKE

A fixed point on two issues. You rank Left, Right and Centre by distance, nearest first.

TWO CAMPS

40% near Left. 60% on the right side, split between Right and Centre.

MAP: WHOM YOU HEAR

A random spot, unrelated to your beliefs. It matters from Experiment 3.

Opinions: 120 voters drawn around (0.25, 0.45), 180 around (0.64, 0.50), spread 0.10 per issue · same recipe in all 30 towns

THE TOWN

The counting rules disagree

118.5
first choices for Left

98
for Right

83
for Centre

HEAD TO HEAD
Centre beats Left **180 to 120** and Right **198 to 102**.

Counting rule	Winner
Plurality: most first choices	Left
Instant runoff	Right (27 of 30)
Borda points	Centre
Beats everyone head to head	Centre

From here on: **plurality, one name per voter**.

Every result is over **30 towns**: 30 random draws of this recipe, and the same 30 in every experiment.

Medians over the 30 towns; ranges: Left 111–128, Right 87–110, Centre 70–99 · Centre wins both head-to-heads in all 30

VOTER RULES

The 4 Types of Voters

Sincere · 40%

“I vote for my favourite.”

Control

Front-runner · 30%

“Only the top two can win. I stay with mine if it's there, else I pick the better of the two.”

Tsang et al. 2018, Top-K

Copycat · 20%

“I vote for the most common vote I see.”

Local majority

Pivot-seeker · 10%

“I switch only if my one vote changes the winner.”

Baumeister et al. 2020

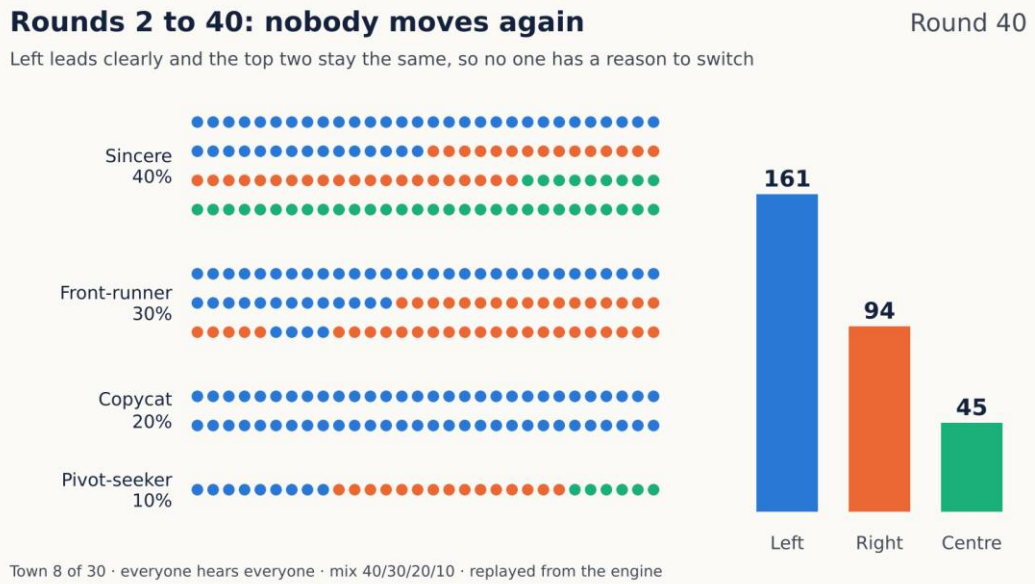
Round 0: everyone votes for their favourite.

Every round after: everyone votes at once, reacting to last round's votes they can see.

Engine policies: sincere, top_k (K = 2), sensed_plurality, pivotal · types are assigned at random, whatever a voter's opinion

EXPERIMENT 1 · THE BASELINE

Everyone hears everyone



Town 8 of 30. Round 0, everyone votes their favourite: Left 120, Right 101, Centre 79. In round 1, 62 voters switch: 161, 94, 45. Then nobody moves.

THE BASELINE

Everyone sees the true count of last round's votes. Default mix 40/30/20/10.

WHAT WE GOT

Frozen after one round in all 30 towns. Left wins all 30; Centre falls from 83 to 41 votes (median).

WHY IT STOPS

Round 1's count has the same top two as round 0, so it points everyone to the vote they already cast. Proved for each of the 30 towns.

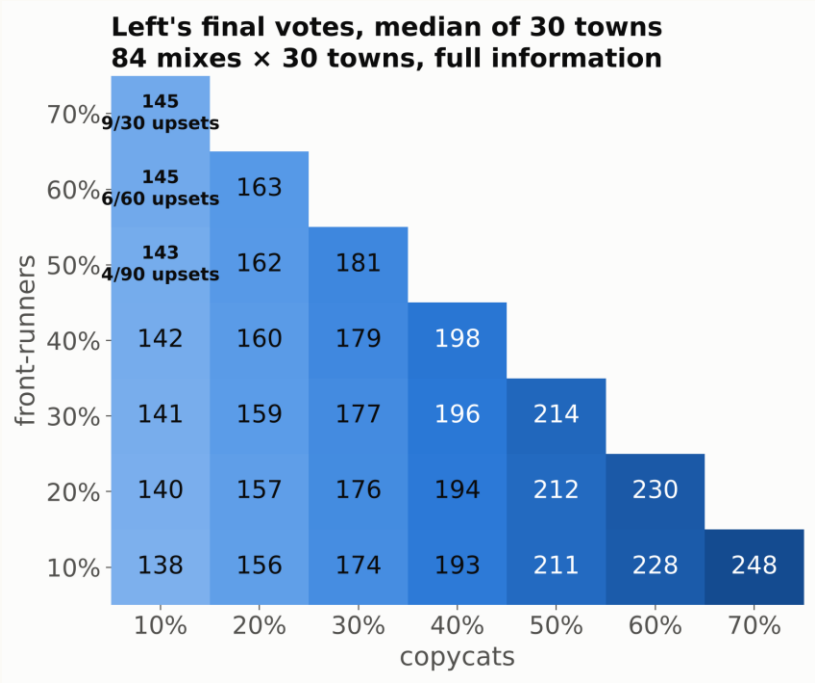
WHY 30 TOWNS

Each town is one random draw of opinions, map spots and voter types. One town could be luck; 30 show the pattern. The same 30 are used in every experiment.

Medians over the 30 towns: Left 118.5 → 159 votes, Right 98 → 98.5, Centre 83 → 41

EXPERIMENT 2

The voter mix



Up: share of front-runners. Across: share of copycats. The rest are sincere voters and pivot-seekers. Each number: Left's final votes, median of 30 towns.

WHAT WE CHANGED

The shares of the four types: all 84 mixes in 10% steps, each type at least 10%. Still full information.

WHAT WE EXPECTED

Different mixes give different winners.

WHAT WE GOT

Left wins 2,501 of 2,520 elections; the 19 upsets go 10 to Right and 9 to Centre. Each extra 10% of copycats adds about 18 votes to Left. Pivot-seekers act like sincere voters.

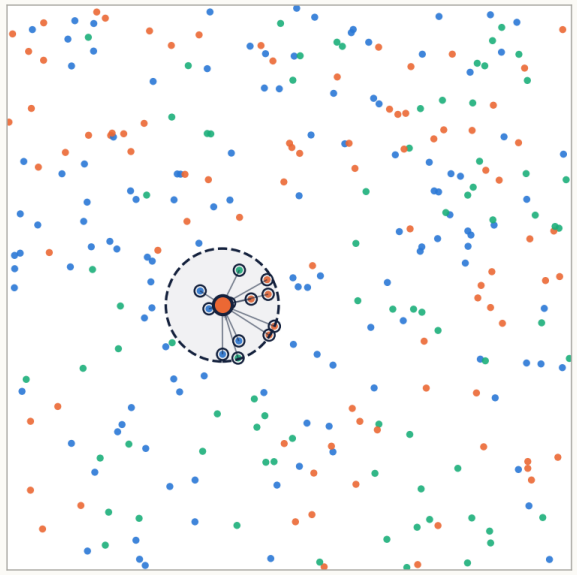
OUR REAL-WORLD READING

With shared information, whether a society has more herders or more tacticians may mostly decide the size of the landslide, not who wins.

Medians: Left 138 → 248 as copycats go 10% → 70% · Centre 66.5 → 16 as strategic voters grow · upsets need 50%+ front-runners

SETUP FOR EXPERIMENT 3

Who you hear: a circle that moves with you

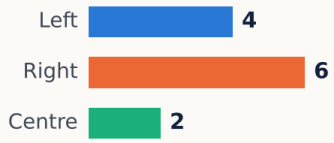


One copycat's hearing circle

Town 8 of 30, radius 0.1, people walking.
This copycat ranks Right > Centre > Left.

Round 3

It hears 12 people's votes from round 2:



Most common: Right
→ it votes Right

Everyone else follows their own rule at the same time.
Replayed from the engine.

Town 8 of 30, radius 0.1, walking, rounds 0 to 12. This copycat ranks Right > Centre > Left, hears 7 to 13 people a round and switches 5 times. Shown: round 3, it hears 12 (Left 4, Right 6, Centre 2) and votes Right.

HEARING RADIUS

You only see last round's votes of people inside your circle. Its size is fixed for a run, and it moves with you.

WALKING

Each round everyone takes a small random step, so the people you hear keep changing.

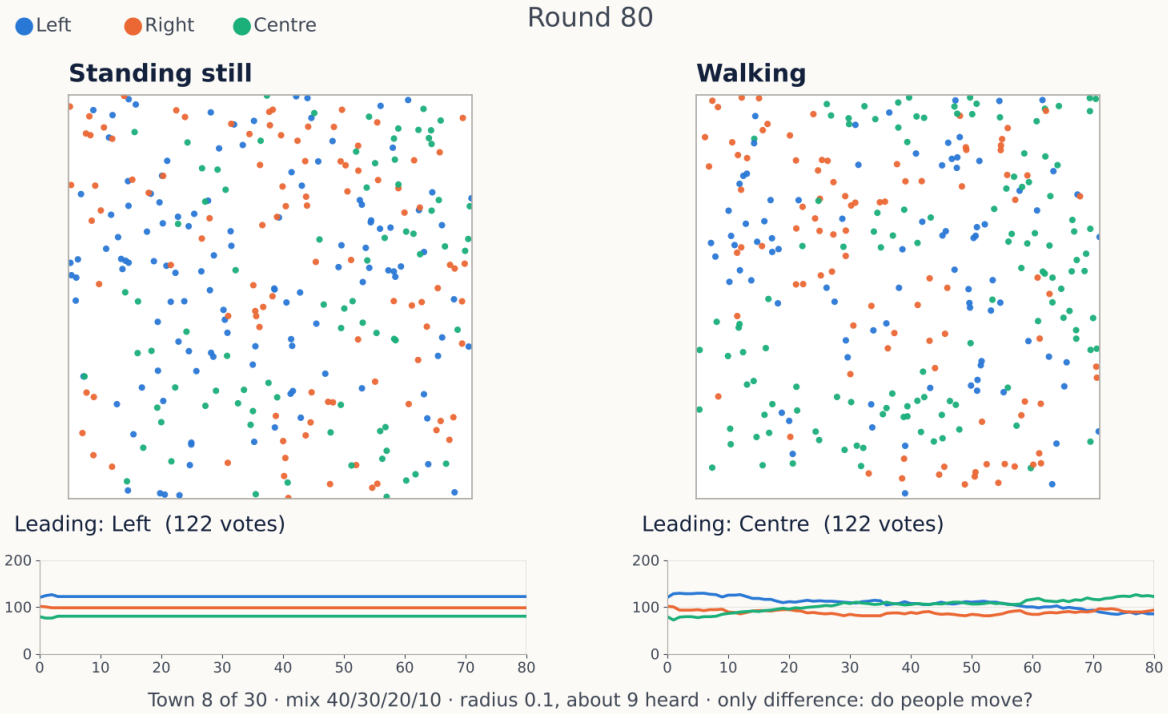
Beliefs stay fixed. Only contacts change.

Radius	People heard
0.4 · 0.2 · 0.1	104 · 32 · 9
0.07 · 0.05	4 · 2

Clip replayed from the engine · each round everyone takes a random step of at most 0.03 on the 1 × 1 map

EXPERIMENT 3 · ONE TOWN

One town, standing still vs walking



Town 8 of 30, mix 40/30/20/10, radius 0.1 (about 9 heard), 80 rounds. The only difference between the two maps: whether people move.

Standing still: Left leads every round. Round 80: Left 122, Right 98, Centre 80. About 6 voters flip back and forth each round.

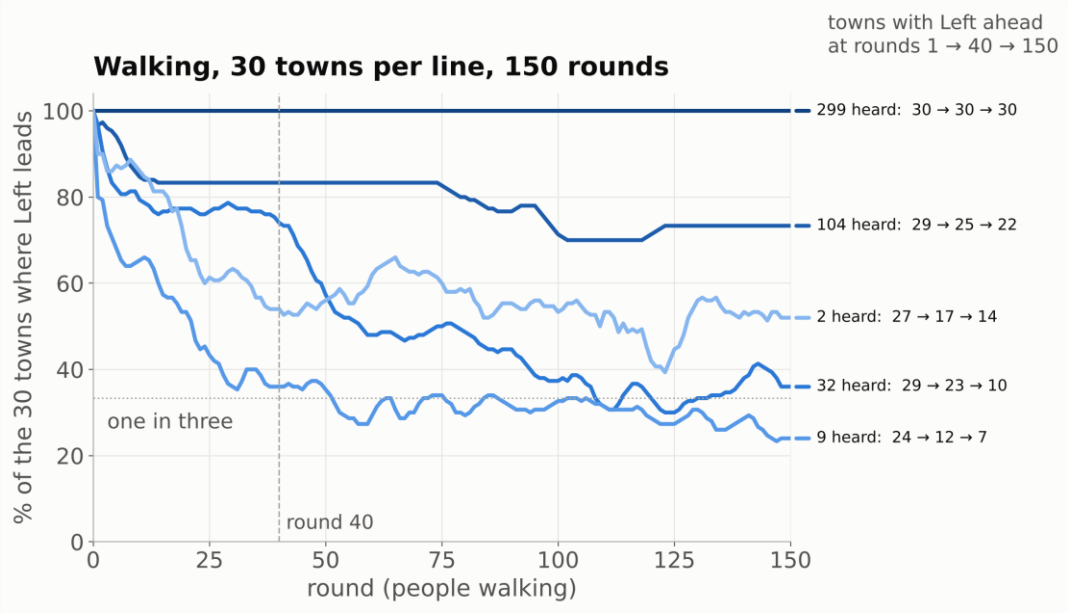
Walking: Left and Centre trade the lead 11 times (rounds 31 to 57). Round 80: Centre 122, Right 93, Left 85. About 11 voters switch each round.

Same people, same opinions. Only whether contacts change.

Both runs replayed from the engine · walking: a random step of at most 0.03 per round on the 1 × 1 map

EXPERIMENT 3

Hearing only neighbours



How to read it: each line is one hearing size, with everyone walking. Its height is the share of the 30 towns where Left, first in round 0 in every town, is still ahead. Right edge: how many towns at rounds 1, 40 and 150.

WHAT WE CHANGED

How many people you hear, from about 104 down to about 2. Walking or standing still.

WHAT WE GOT

Standing still, towns settle; Left wins 23 to 26 of 30. Walking with 32 heard or fewer, votes never settle. With 32 heard, Left leads in 29 of 30 towns after round 1, 10 by round 150.

WHY

Your circle is a small sample, and nearby Right or Centre may be ahead. Copycats and front-runners react to that local picture, so Left's town-wide bandwagon never fully forms. Walking changes the sample every round, so people keep re-deciding.

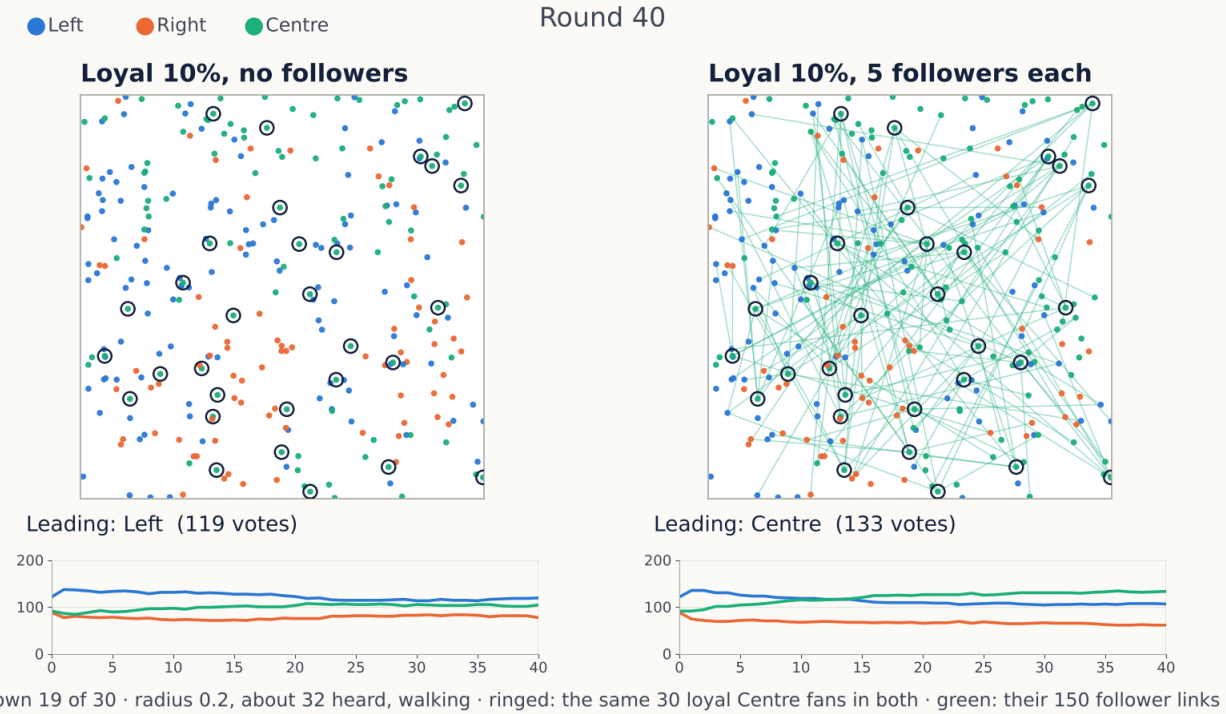
OUR REAL-WORLD READING

Each round is an election every four years: if circles reshuffle, the same voters with the same beliefs can elect different winners.

Engine detail: front-runners are sticky with 300 voters; with 301, votes still never settle and Left wins 20–28 of 30

SETUP FOR EXPERIMENT 4

An organised minority with an audience



Town 19 of 30, radius 0.2, walking, 40 rounds. Both maps have the same 30 loyal Centre fans, ringed. Only on the right do they have followers: 5 each, the green lines.

30 Centre fans, 10% of the town, always vote Centre.

Each has **5 followers**: random voters anywhere in town who see that member's vote every round, like a social media audience.

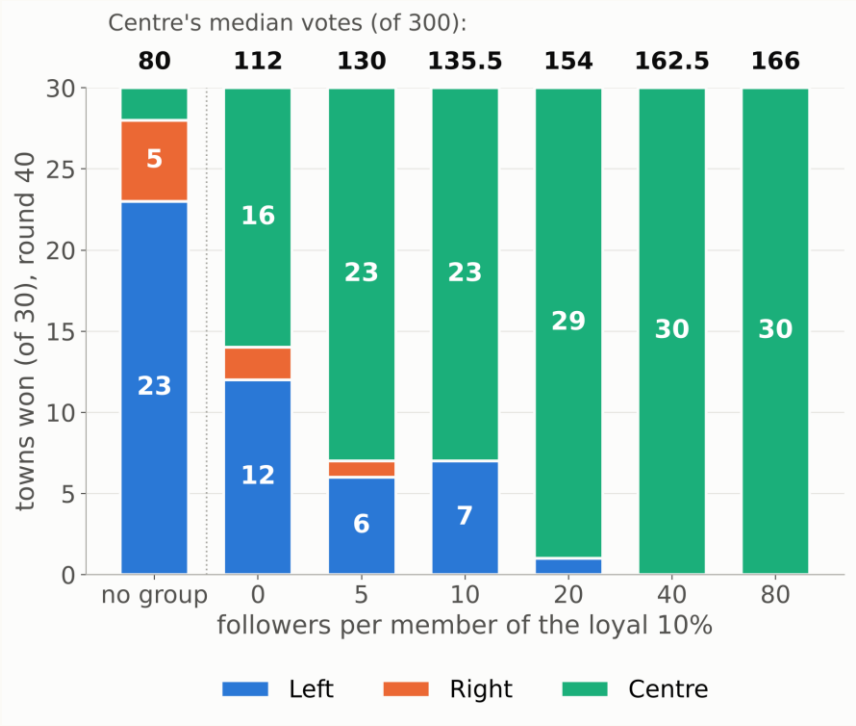
150 links: about **2 in 5 voters** also see a loyal fan's vote.

Everyone else hears about 32 neighbours and walks.

Round 40 in town 19 · no followers: Left 119, Centre 104, Right 77 · 5 followers each: Centre 133, Left 106, Right 61

EXPERIMENT 4

Propaganda



30 towns per bar, round 40, radius 0.2, walking. On top: Centre's median votes. First bar: no group.

WHAT WE CHANGED

First loyalty: 30 Centre fans who never waver. Then reach: 0 to 80 followers per member.

WHAT WE GOT

Loyalty alone: Centre wins 16 of 30 towns, up from 2. Just 5 followers each: 23. With 20: 29, and it holds over 150 rounds.

WHY IT WORKS

Centre is almost everyone's second choice: wherever it looks like one of the top two nearby, tactical voters switch to it, and where it leads, copycats follow. Loyal fans keep it in that top two; one extra Centre vote, seen by a follower, can tip a close local count.

OUR REAL-WORLD READING

Being heard, not hearing, is the lever: a small group with an audience could swing an election without changing anyone's beliefs.

Radius 0.2, about 32 heard, walking, round 40 · with 301 voters: 13, 15, 24 and 30 of 30 for 0, 5, 20 and 40 followers

MULTI-AGENT ELECTION LAB · PART 2

Who steers the winner on purpose

Four deliberate manipulations, each run against an untouched twin. Who wins, how, and who can catch them.

Team Eclipse · Experiments 5 to 8 · towns of 100 people, same engine

THE QUESTION

If someone tries to steer the result on purpose, does it work, and can anyone catch them?

EXPERIMENT 5

A watchdog that sees only totals

10 attack styles · 420 runs

EXPERIMENT 6

Foreign spies

15 sleeper cells · 4 tactics × 8 towns

EXPERIMENT 7

Changing minds slowly

4 campaigns, same budget × 5 towns

EXPERIMENT 8

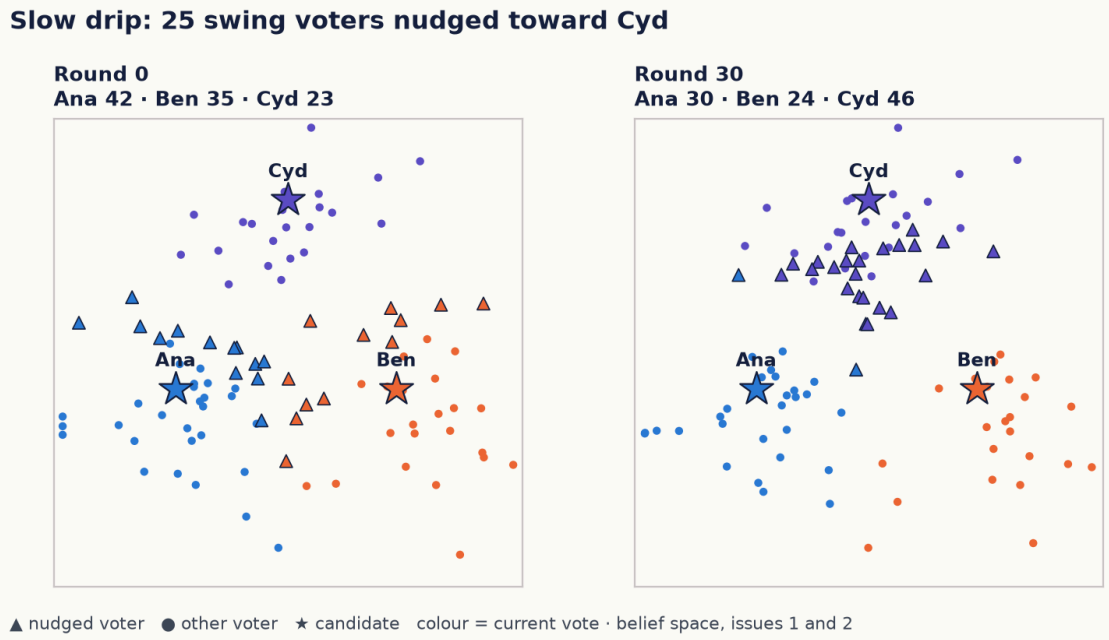
Campaigns vs echo chambers

3 parties × 5 playbooks: 125 match-ups × 3 towns

Every attack run is paired with a twin town that is identical except that the attacker stays quiet

SETUP FOR EXPERIMENT 5

A watchdog that sees only the scoreboard



Attack W4, slow drip. Every round the attacker nudges the beliefs of the 25 voters closest to switching a little toward Cyd. By round 30 Cyd has gone from third (23) to first (46).

THE TOWN

100 people with beliefs on two issues. Three candidates: **Ana 42, Ben 35, Cyd 23**. Everyone votes for the nearest candidate; beliefs also drift a little by themselves.

THE ATTACKER

A fixed budget to push chosen voters' beliefs toward Cyd. Ten styles: fast or slow, few or many voters, aimed or random.

THE WATCHDOG

Sees only each round's vote totals. It **wins** if it raises the alarm before Cyd wins three rounds in a row.

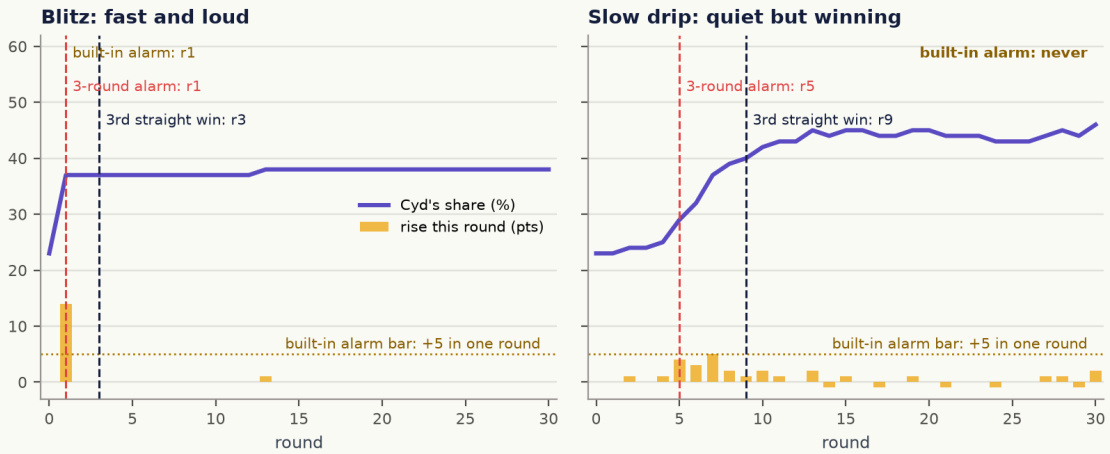
BUILT-IN ALARM

Fires when the target jumps more than 5 points in one round.

Belief drift up to 0.01 per round · 30 rounds · every attack paired with an untouched twin town

EXPERIMENT 5

Slow attacks slip under a one-round alarm



Purple: Cyd's share. **Yellow bars:** its rise each round. A blitz jumps 14 points at once; the slow drip never rises more than 5 in a round, yet wins by round 9.

WHAT WE CHANGED

Ten ways to spend the attack budget, one town each, each with an untouched twin.

WHAT WE GOT

The built-in alarm catches all 3 fast attacks, but **misses both slow drips**, which still win by round 9 and 10. Five attacks never win: wrong targets, random pushes, too little money, too few or too many voters.

WHY

A drip adds 4 to 5 points a round: always under the one-round bar, but it keeps adding up.

OUR REAL-WORLD READING

Watching for sudden jumps misses patient campaigns.

No alarm fired in any of the 10 twin towns · the attack only wins on the 25 voters closest to switching

EXPERIMENT 5 · BEST STRATEGIES

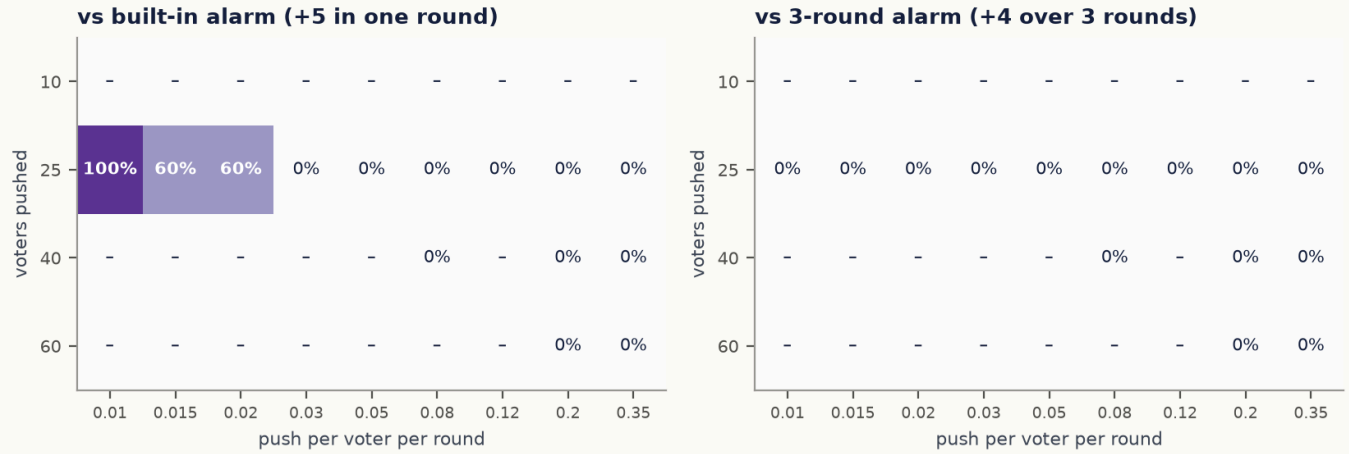
Look back three rounds, not one

20 of 25
attacks caught by the built-in alarm

25 of 25
caught by the 3-round alarm

0 of 40
false alarms in untouched towns

0 of 36
attack plans that win unseen



Cell: share of 5 test towns where the attacker wins without being caught first · – = attack never wins

THE WATCHDOG'S BEST RULE

We tried 101 rules that use only vote totals and kept the fastest one that never fired in 80 untouched towns, even with 50% more natural drift: **alarm if the target rose more than 4 points over the last 3 rounds**.

THE ATTACKER'S BEST REPLY

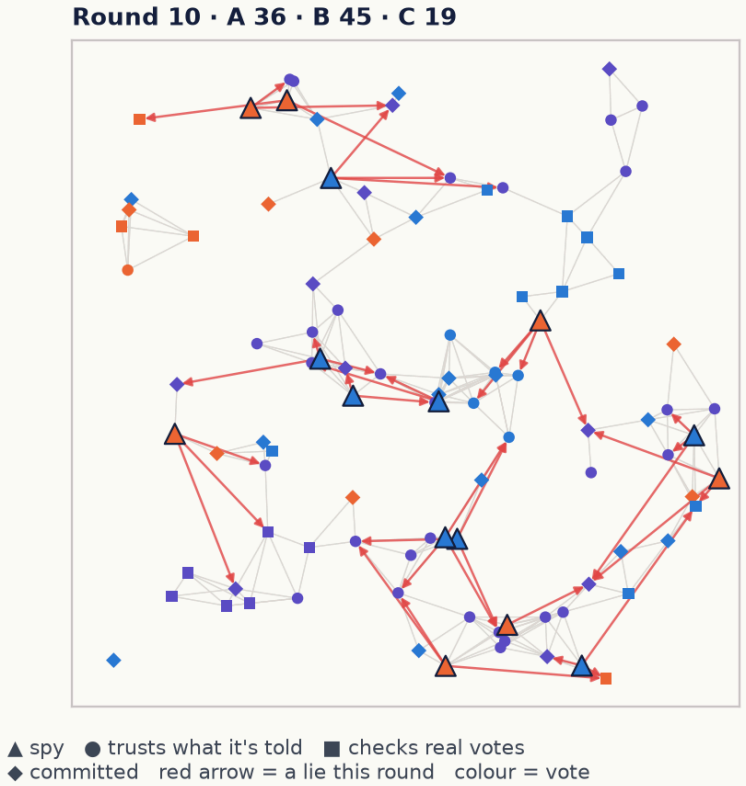
Against the one-round alarm: a slow drip on the 25 swing voters wins **every time, unseen**, around round 17. Against the 3-round rule nothing slips through; the best left is a blitz that is flagged in round 1.

Rule chosen on towns 201–205, tested on new towns 101–105 · lowering the one-round bar to 3 points also catches 25 of 25

SETUP FOR EXPERIMENT 6

Sleeper cells in the group chats

Spy cells with a dossier, one town



THE TOWN

85 residents: **A 40%, B 37%, C 23%**, chatting with neighbours and in 10 group chats. Three kinds of resident:

- ◆ **Committed** 40%: always vote their favourite.
- ● **Trusting** 40%: vote what most contacts say they vote.
- ■ **Checkers** 20%: vote what most neighbours actually vote.

THE SPIES · 15%

15 sleepers (▲) vote like locals to blend in and send fake “I'm voting B” messages: 4 a round, 100 each, to their 15 followers and one group chat.

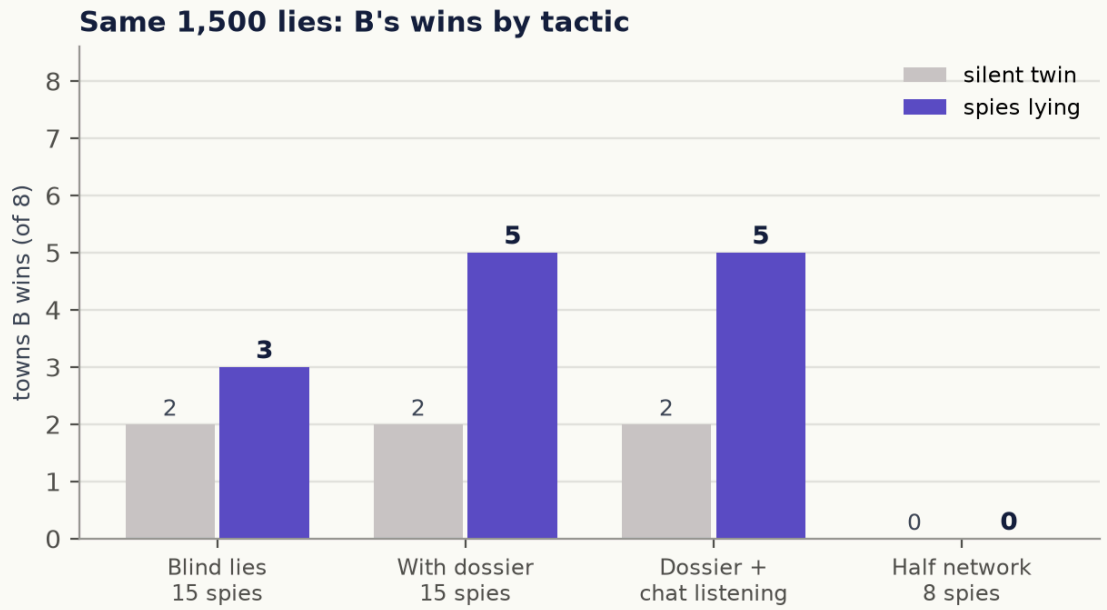
THEIR INTELLIGENCE

None (lie at random), a **dossier** of their followers' true favourites, or the dossier plus **listening** to two nearby group chats.

Each spy is its own cell with its own budget · 25 rounds · every run paired with a twin whose spies stay silent

EXPERIMENT 6

Knowing whom to lie to is the weapon



8 towns per tactic. Grey: the twin town where the spies stay silent. Purple: the same town with the spies lying. Every tactic sends the same 1,500 lies.

WHAT WE CHANGED

Only what the spies know: nothing, a dossier, dossier plus chat listening; then half the network.

WHAT WE GOT

B wins 2 of 8 towns with silent spies, **3 with blind lies, 5 with a dossier**. Listening to chats adds nothing. With 8 spies B never leads.

WHY

Trusting voters break ties with their own favourite, so hidden B fans who now follow A tip with one or two lies. Blind lies land on committed voters and checkers, who never move.

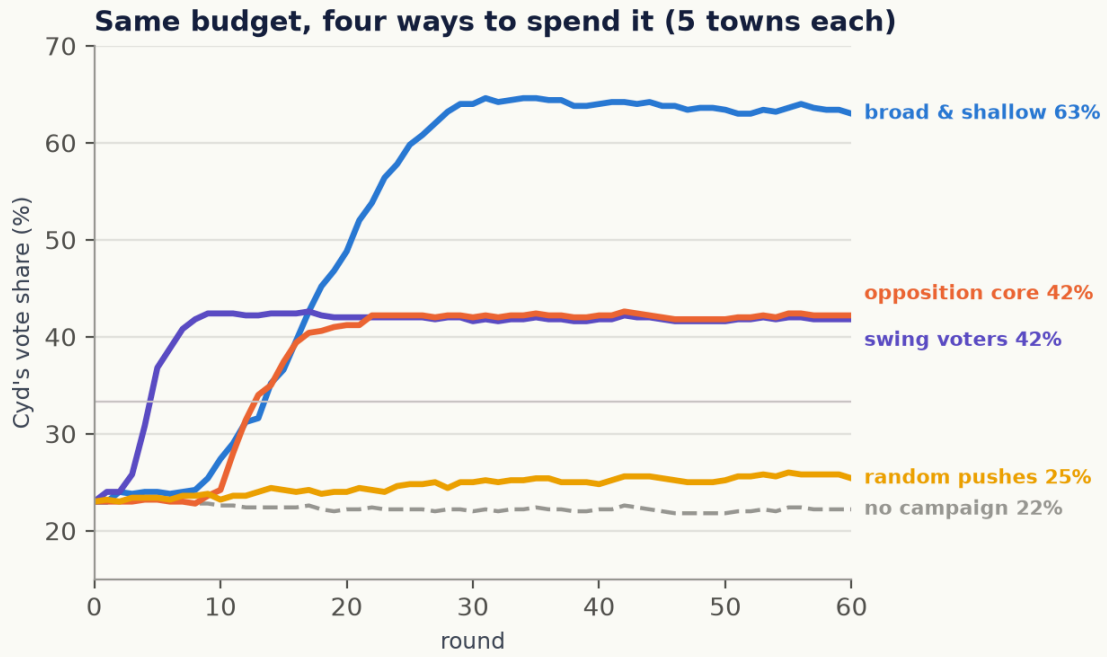
OUR REAL-WORLD READING

Targeting data, not volume, is what makes disinformation work; voters who check what people do, not what they say, are nearly immune.

B's share gain over the silent twin: blind +2.0, dossier +4.8 points · the 8-spy town also starts weaker for B (23% vs 36% in its twin)

EXPERIMENT 7

Changing minds slowly



Same town as Experiment 5, 60 rounds. Each campaign gets the same budget for pushing beliefs toward Cyd and spends it in a different way. Mean of 5 towns.

WHAT WE CHANGED

Where the budget goes: a little on 60 voters, a lot on the 20 swing voters, on 20 of the opposition's core, or in random directions.

WHAT WE GOT

Swing voters: Cyd leads by round 5 and wins 56 of 61 rounds.
Broad and shallow: slower, but 63% of the vote. The core works slowly; random pushes do nothing. The gains stay after the money stops.

WHY

Beliefs drift at random with no pull back, so moved voters stay moved.

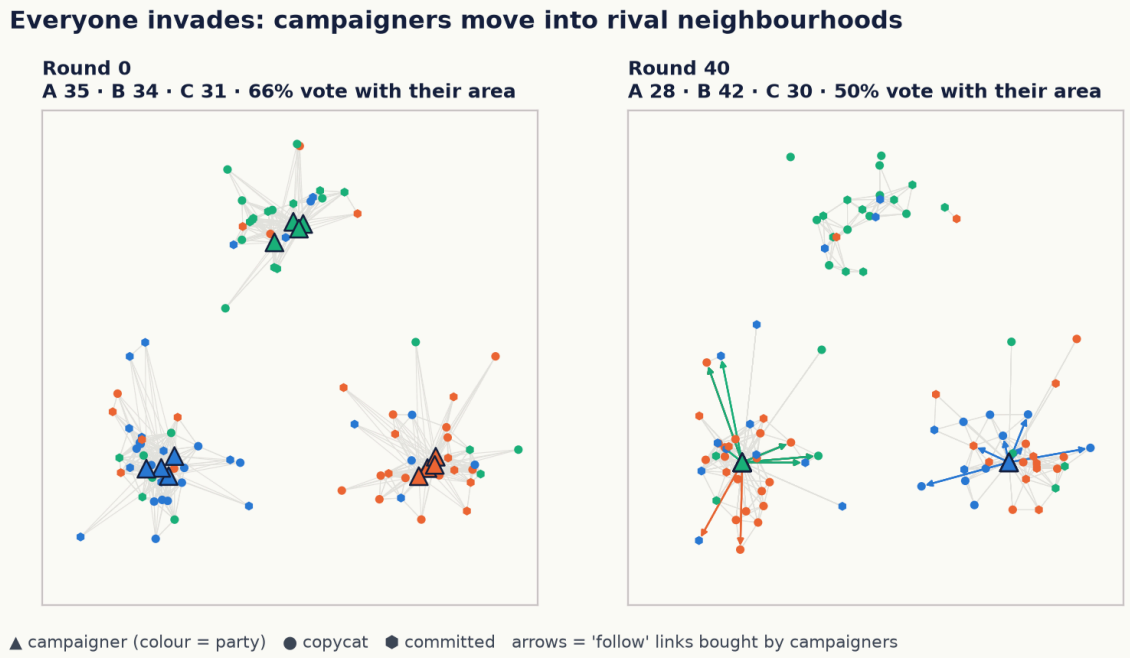
OUR REAL-WORLD READING

The winner ends up 24% further from what people first believed, yet looks closer to what they believe now: successful persuasion hides itself.

Average distance from the winner to voters' original beliefs: 0.436 with any successful campaign vs 0.352 without one

SETUP FOR EXPERIMENT 8

Campaigners who reshape who you hear



All three parties invade. Campaigners walk into rival neighbourhoods and make voters follow them. By round 40 only 50% still vote with their neighbourhood, and B has taken the lead.

THE TOWN

88 residents in three neighbourhoods: **A 34, B 30, C 24**, 65% loyal to the local party. Copycats (60%) hear only 4 or 5 neighbours; the rest are committed.

THE CAMPAIGNERS

4 per party. They walk 6× faster, poll everyone within 0.3, and can make up to 6 voters follow them, 2 new a round.

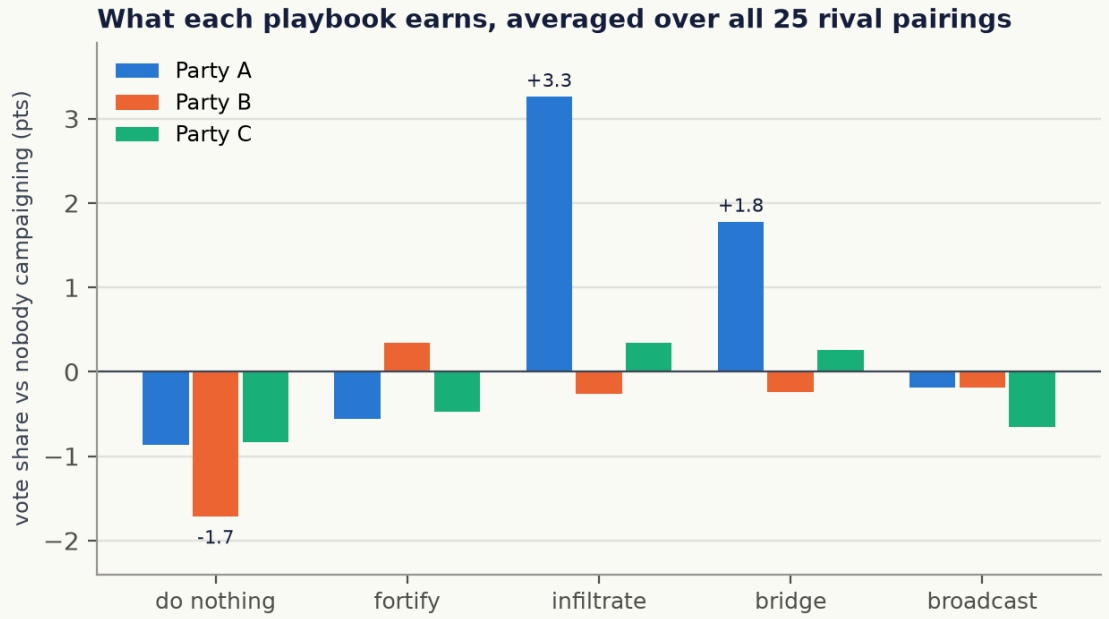
FIVE PLAYBOOKS

Do nothing · **fortify**: stay home, win back defectors · **infiltrate**: move into the biggest rival area · **bridge**: sit on the border, chase swing voters · **broadcast**: random followers.

Every combination of playbooks, 125 in all, × 3 towns × 40 rounds · a party's score: its share over the last 10 rounds

EXPERIMENT 8

The leader attacks, the challenger defends



Each bar: a party's average gain from one playbook, over all 25 combinations of its rivals' playbooks.

WHAT WE CHANGED

Every party's playbook, in every combination.

WHAT WE GOT

A should infiltrate: best in 19 of 25 match-ups, +3.3 points. **B** should fortify; doing nothing costs it 1.7. **C** barely moves: +0.3 at best, and wins **0 of 375** elections.

WHY

Invading the biggest rival splits its copycats. The smallest party has no area big enough to tip.

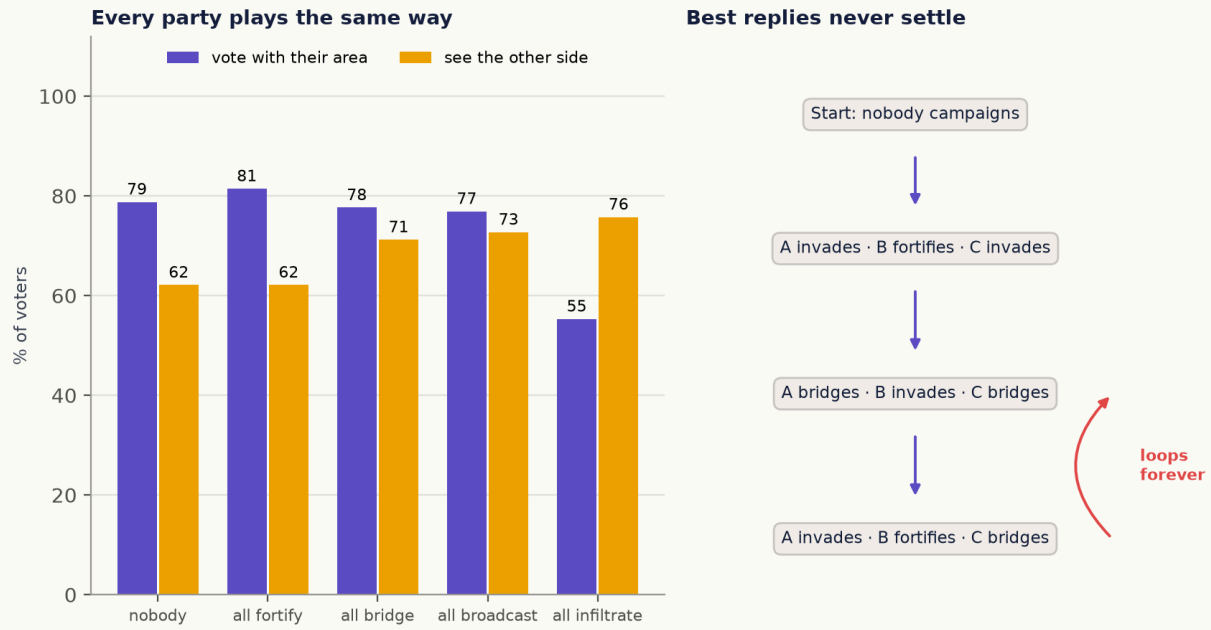
OUR REAL-WORLD READING

B's one route to an upset is to work the border while A stays home: then it wins 2 of 3 towns.

Across all 375 runs: A wins 348, B 27, C 0 · averages over 3 towns per match-up

EXPERIMENT 8 · BEST STRATEGIES

No resting point, only an arms race



Left: every party plays the same playbook. **Right:** each party in turn switches to its best reply; the switching never stops.

WHAT WE EXPECTED

One set of playbooks where no party wants to switch.

WHAT WE GOT

There is none. The best replies loop forever: A swaps invading for bridging, B swaps fortifying for invading. Closest to rest: **everyone invades**; no party gains even a quarter of a vote by switching.

OUR REAL-WORLD READING

When every side campaigns in its rivals' areas, echo chambers break (79% → 55% vote with their neighbourhood; 62% → 76% see the other side), but the winner barely changes.

Most stable mixes: A bridges, B and C invade (best switch gains 0.20 points) and everyone invades (0.23)

TAKEAWAYS

1. With shared information the first leader locks in: in all 30 towns after one round, proved for each.

2. The mix of voters moves the margin, almost never the winner: 19 upsets in 2,520 elections.

3. Changing contacts make the winner unstable, even with fixed beliefs: with 32 heard, Left leads in 29 of 30 towns after round 1, 10 by round 150.

4. A loyal minority with even a small audience can decide the result: with 5 followers each, Centre goes from 2 to 23 of 30 towns.

TAKEAWAYS · PART 2

5. A one-round alarm misses patient attackers: a 3-round look-back caught 25 of 25, with no false alarms.

6. Targeting beats volume: the same lies take B from 2 to 5 of 8 towns once spies know whom to lie to.

7. Slow persuasion sticks and hides itself: the winner ends up 24% further from what people first believed.

8. Campaign wars break echo chambers but not the winner: no stable playbook, and the underdog won 0 of 375.

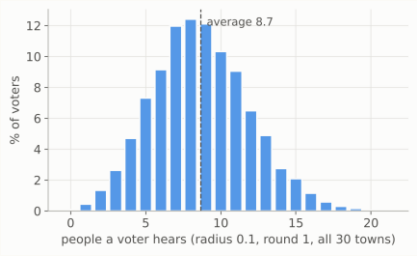
POSSIBLE EXTENSIONS

Where this could go next

Mixed voter types

Each voter a blend, say 60% sincere and 40% front-runner, that can shift with how close the race is.

Unequal hearing



Same radius, 0 to 21 people heard. Hubs, echo chambers.

Ballots and rules

- Ranked ballots: instant runoff, Borda
- More candidates

People

- Realistic opinion distributions
- Opinions that drift
- Front-runner stickiness as a setting